Mathematics Preliminaries for Machine Learning

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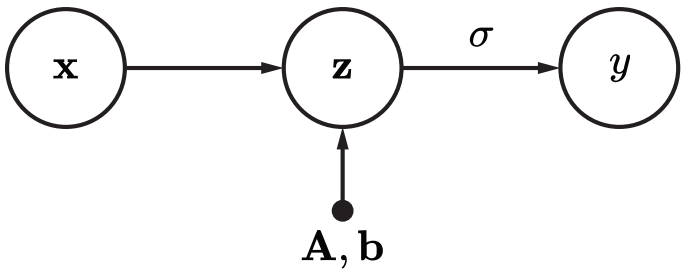
Motivation

Motivation

- Mathematics tools required:
  - linear algebra,
  - calculus,
  - optimization,
  - probability,
  - information theory.
- For example, a machine learning model can be given:



$$
\begin{aligned}
\mathbf{z} &= \mathbf{A}\mathbf{x} + \mathbf{b} \\
y &= \sigma(\mathbf{z}) \\
&= \sigma(\mathbf{A}\mathbf{x} + \mathbf{b}).
\end{aligned}
$$

Notation

Table 1: Notation.

Notation	Description
$\mathbb{R}$	set of real numbers
$\mathbb{R}^n$	set (vector space) of n-tuples of real numbers
$\mathbb{R}^{m\times n}$	set (vector space) of m-by-n matrices
$\mathbf{x}$	a vector $\in \mathbb{R}^n$
$\mathbf{A}$	a matrix $\in \mathbb{R}^{m\times n}$
$\nabla f(\mathbf{x})$	gradient of the function f at $\mathbf{x}$
$\nabla^2 f(\mathbf{x})$	Hessian of the function f at $\mathbf{x}$

Table 2: Notation.

Notation	Description
Ω	sample space
$\mathbb{P}(A)$	probability of event A
$p(X)$	distribution of random variable X
$p(x)$	probability density/mass function evaluated at x
A^c	complement of event A
$A \cup B$	union of A and B
$\mathbb{E}[X]$	expected value of random variable X
$\mathrm{Var}(X)$	variance of random variable X
$\mathrm{Cov}(X, Y)$	covariance of random variables X and Y

Linear Algebra

Linear Algebra

- Scalars, vectors, matrices, and tensors
- Vector spaces and linear maps
- Metric spaces and normed spaces
- Transpose and matrix multiplication
- Identity and inverse matrices
- Diagonal, symmetric, and orthogonal
- Linear independence and rank
- Eigenthings and eigendecomposition
- Range space and null space
- Determinant and trace
- Singular value decomposition
- Positive (semi-)definite matrices

Scalars, Vectors, Matrices, and Tensors

- A scalar is a single number, $x \in \mathbb{R}$.
- A vector is an array of numbers,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n.$$

- A matrix is a two-dimensional array of numbers,

$$\mathbf{x} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & & & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

- A tensor is a generalized version of high-dimensional array of numbers.

Vector Spaces

- A vector space V is a set on which two operations are defined: vectors can be added together and vectors can be multipled by real numbers.
- V must satisfy
  1. There exists an additive identity (written $\mathbf{0}$) in V such that $\mathbf{x} + \mathbf{0} = \mathbf{x}$ for all $x \in V$
  2. For each $\mathbf{x} \in V$, there exists an additive inverse (written $-\mathbf{x}$) such that $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$
  3. There exists a multiplicative identity (written 1) in $\mathbb{R}$ such that $1\mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in V$
  4. Commutativity, associativity and distributivity.

Linear Maps

- A linear map is a function $T : V \to W$, where V and W are vector spaces, that satisfies
  1. $T(\mathbf{x} + \mathbf{y}) = T\mathbf{x} + T\mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in V$
  2. $T(\alpha\mathbf{x}) = \alpha T\mathbf{x}$ for all $\mathbf{x} \in V$, $\alpha \in \mathbb{R}$.
- It is also called as a linear mapping, linear transformation, or sometimes linear function.

Metric Spaces and Normed Spaces

- A metric on a set S is a function $d : S \times S \to \mathbb{R}$ that satisfies
  1. $d(x, y) \geq 0$, with equality if and only if $x = y$
  2. $d(x, y) = d(y, x)$
  3. $d(x, z) \leq d(x, y) + d(y, z)$

  for all $x, y, z \in S$.
- A norm on a real vector space V is a function $\| \cdot \| : V \to \mathbb{R}$ that satisfies
  1. $\|\mathbf{x}\| \geq 0$, with equality if and only if $\mathbf{x} = \mathbf{0}$
  2. $\|\alpha \mathbf{x}\| = |\alpha| \|\mathbf{x}\|$
  3. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$

  for all $\mathbf{x}, \mathbf{y} \in V$ and all $\alpha \in \mathbb{R}$.
- Specific norms on $\mathbb{R}^n$ can be defined:

$$\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i|, \quad \|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}, \quad \|\mathbf{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$$

where $p \geq 1$.

Transpose and Matrix Multiplication

- The transpose of $\mathbf{A}$ is $\mathbf{A}^\top \in \mathbb{R}^{n\times m}$ where $[\mathbf{A}^\top]_{ij} = A_{ji}$.
- The dot product (inner product) between two vectors $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^n$ is $\mathbf{x}^\top\mathbf{y} = \sum_{i=1}^n x_i y_i$.
- Given $\mathbf{A} \in \mathbb{R}^{m\times k}$ and $\mathbf{B} \in \mathbb{R}^{k\times n}$, a matrix multiplication $\mathbf{AB} = \mathbf{C} \in \mathbb{R}^{m\times n}$ is $C_{ij} = \sum_{l=1}^k A_{il}B_{lj} = \mathbf{a}_{i:}\mathbf{b}_{:j}$.
- Given $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{B} \in \mathbb{R}^{m\times n}$, Hadamard product $\mathbf{A} \odot \mathbf{B} = \mathbf{C} \in \mathbb{R}^{m\times n}$ is $C_{ij} = A_{ij}B_{ij}$.
- The outer product between two vectors $\mathbf{x} \in \mathbb{R}^m$ and $\mathbf{y} \in \mathbb{R}^n$ is $\mathbf{x} \otimes \mathbf{y} = \mathbf{x}\mathbf{y}^\top \in \mathbb{R}^{m\times n}$.

Identity and Inverse Matrices

- Identity matrix $\mathbf{I} \in \mathbb{R}^{n \times n}$ is

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

- The matrix inverse of $\mathbf{A}$ satisfies $\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

Diagonal, Symmetric, and Orthogonal

- A matrix $\mathbf{D}$ is diagonal if and only if $D_{ij} = 0$ for all $i \neq j$.
- A matrix $\mathbf{A}$ is symmetric if and only if $\mathbf{A} = \mathbf{A}^\top$.
- A matrix $\mathbf{A}$ is orthogonal if and only if $\mathbf{A}\mathbf{A}^\top = \mathbf{A}^\top\mathbf{A} = \mathbf{I}$.

Linear Independence and Rank

- The vectors $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ are said to be linearly independent if the equation
$$a_1\mathbf{u}_1 + \cdots + a_n\mathbf{u}_n = \mathbf{0}$$
is only satisfied by $a_1 = \cdots = a_n = 0$.
- The rank of a matrix $\mathbf{A}$ is the maximal number of linear independent columns (or rows) of $\mathbf{A}$.

Eigenthings and Eigendecomposition

- For a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, a nonzero vector $\mathbf{x} \in \mathbb{R}^n$ is an eigenvector of $\mathbf{A}$ corresponding to eigenvalue λ if

$$\mathbf{A}\mathbf{x} = \lambda \mathbf{x}.$$

- The zero vector is excluded from this definition because $\mathbf{A}\mathbf{0} = \mathbf{0} = \lambda \mathbf{0}$ for every λ.
- Suppose that the matrix $\mathbf{A}$ has full rank. Then the eigendecomposition of $\mathbf{A}$ is given by

$$\mathbf{A} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^{-1}$$

where $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_n]$ and $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

Range Space and Null Space

- For any $\mathbf{A} \in \mathbb{R}^{m \times n}$, its range space $\mathrm{Range}(\mathbf{A})$ and null space $\mathrm{Null}(\mathbf{A})$ are defined:

$$\mathrm{Range}(\mathbf{A}) = \{\mathbf{y} \in \mathbb{R}^m | \mathbf{y} = \mathbf{A}\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n$$
$$\mathrm{Null}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{R}^n | \mathbf{A}\mathbf{x} = \mathbf{0}\}.$$

Determinant and Trace

- The determinant of a square matrix $\det(\mathbf{A})$ or $|\mathbf{A}|$ is

$$\det(\mathbf{A}) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

- The trace of a square matrix is the sum of its diagonal entries:

$$\mathrm{tr}(\mathbf{A}) = \sum_{i=1}^{n} A_{ii}.$$

Singular Value Decomposition

- Singular value decomposition is a matrix decomposition for every matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

where $\mathbf{U} \in \mathbb{R}^{m \times m}$ and $\mathbf{V} \in \mathbb{R}^{n \times n}$ are orthogonal matrices and $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$ a diagonal matrix with the singular values of $\mathbf{A}$ (denoted as σ_i) on its diagonal.

Positive (Semi-)Definite Matrices

- A symmetric matrix $\mathbf{A}$ is positive semi-definite if for all $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^\top \mathbf{A}\mathbf{x} \geq 0$.
- A symmetric matrix $\mathbf{A}$ is positive definite if for all nonzero $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^\top \mathbf{A}\mathbf{x} > 0$.
- A symmetric matrix is positive semi-definite if and only if all of its eigenvalues are nonnegative, and positive definite if and only if all of its eigenvalues are positive.
- Suppose $\mathbf{A} \in \mathbb{R}^{m \times n}$. Then $\mathbf{A}^\top \mathbf{A}$ is positive semi-definite. If $\mathrm{Null}(\mathbf{A}) = \{\mathbf{0}\}$, then $\mathbf{A}^\top \mathbf{A}$ is positive definite.

Probability

Probability

- Probability in machine learning
- Sets, fields, and events
- Probability measure
- Probability space
- Joint, marginal, conditional, total probabilities and independence
- Chain rule and Bayes' rule
- Random variables
- Probability distribution and density
- Expectations, variance, and covariance
- Random vectors
- Gaussian distribution and KL divergence

Probability in Machine Learning

- Machine learning deals with uncertain quantities or stochastic quantities:
  - inherent stochasticity in the system being modeled
  - incomplete observability
  - incomplete modeling.
- A probabilistic machine learning model is a joint distribution $p(\mathbf{x}, \mathbf{z})$, over observed variables $\mathbf{x}$ and hidden variables $\mathbf{z}$.
- Inference about unknowns is carried out by calculating the posterior distributions over hidden variables:

$$p(\mathbf{z}|\mathbf{x}) = \frac{p(\mathbf{x}, \mathbf{z})}{p(\mathbf{x})}.$$

Sets, Fields, and Events

- A set is a collection of objects. The objects are called elements of the set.
- A sample space Ω is the set of all outcomes of an experiment.
- A subset of Ω is called event. A collection of subsets of Ω are called events.
- Consider a universal set Ω and a collection of subsets of Ω. Let $E, F, \ldots$ denote subsets in this collection. This collection of subsets of Ω forms a field $\mathcal{M}$ if
  1. $\emptyset \in \mathcal{M}$, $\Omega \in \mathcal{M}$
  2. if $E \in \mathcal{M}$ and $F \in \mathcal{M}$, then $E \cup F \in \mathcal{M}$, and $E \cap F \in \mathcal{M}$
  3. if $E \in \mathcal{M}$, then $E^c \in \mathcal{M}$.
- A σ-field $\mathcal{F}$ is a field that is closed under any countable set of unions, intersections, and combinations.

Probability Measure

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- A σ-field $\mathcal{F}$ is a field that is closed under any countable set of unions, intersections, and combinations.

Joint, Marginal, Conditional, Total Probabilities and Independence

- Joint probability is $\mathbb{P}(A, B)$.
- Marginal probability is $\mathbb{P}(A) = \sum_B \mathbb{P}(A, B)$.
- Conditional probability is $\mathbb{P}(A|B) = \frac{\mathbb{P}(A,B)}{\mathbb{P}(B)}$.
- Total probability is $\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B|A_i)\mathbb{P}(A_i)$ where $\cup_{i=1}^{n} A_i = \Omega$.
- Independence is $\mathbb{P}(A, B) = \mathbb{P}(A)\mathbb{P}(B)$.

Chain Rule and Bayes' Rule

- Chain rule is

$$\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B) = \mathbb{P}(B|A)\mathbb{P}(A).$$

- Bayes' rule is

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Random Variables

- A random variable is some uncertain quantity with an associated probability distribution over the values it can assume.
- A random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a function $X : \Omega \to \mathbb{R}$.
- We denote the range of X by $X(\Omega) = \{X(w) : w \in \Omega\}$.
- For example, suppose X is the number of heads in two tosses of a fair coin. The sample space is

$$\Omega = \{hh, tt, ht, th\}$$

and X is determined completely by the outcome w.

Probability Distribution and Density

- Discrete random variables
  - Probability mass function is $p(x) = \mathbb{P}(X = x)$.
- Continuous random variables
  - Probability density function is $p(x) = \mathbb{P}(X = x)$.
  - Cumulative distribution function is

$$F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} p(z) \, \mathrm{d}z.$$

Expectations, Variance, and Covariance

- Expectation $\mathbb{E}[X]$ is $\mathbb{E}[X] = \sum_{x \in X(\Omega)} xp(x)$ for discrete X and $\mathbb{E}[X] = \int_{-\infty}^{\infty} xp(x)\,\mathrm{d}x$ for continuous X.
- Variance $\mathrm{Var}(X)$ is $\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$.
- Covariance $\mathrm{Cov}(X, Y)$ is $\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$.

Random Vectors

- Multivariate distributions is defined as distributions of random vectors,

$$\mathbf{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix}.$$

- Expectation of a random vector is

$$\mathbb{E}[\mathbf{X}] = \begin{bmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{bmatrix}.$$

Random Vectors

- The variance is generalized by the covariance matrix:

$$
\begin{aligned}
\mathbf{\Sigma} &= \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top] \\
&= \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots & \mathrm{Cov}(X_1, X_n) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots & \mathrm{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_n, X_1) & \mathrm{Cov}(X_n, X_2) & \cdots & \mathrm{Var}(X_n) \end{bmatrix}.
\end{aligned}
$$

Gaussian Distribution

- Univariate Gaussian distribution is

$$p(x) = \mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x-\mu)^2\right\}.$$

- Multivariate Gaussian distribution is

$$\begin{aligned} p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \\ &= \frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}. \end{aligned}$$

KL Divergence

- A measure of how one probability distribution q diverges from the other p.
- KL divergence $D_{\mathrm{KL}}[p\|q]$ is

$$D_{\mathrm{KL}}[p\|q] = \sum_{\mathbf{x}} p(\mathbf{x}) \log \frac{p(\mathbf{x})}{q(\mathbf{x})}$$

for discrete probability distributions p and q, or

$$D_{\mathrm{KL}}[p\|q] = \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{q(\mathbf{x})} \, \mathrm{d}\mathbf{x}$$

for continuous probability distributions p and q.

KL Divergence of Two Gaussians

- Two univariate Gaussians $p(x) = \mathcal{N}(\mu_1, \sigma_1^2)$ and $q(x) = \mathcal{N}(\mu_2, \sigma_2^2)$,

$$D_{\mathrm{KL}}[p\|q] = \frac{1}{2}\frac{\sigma_1^2}{\sigma_2^2} + \frac{1}{2}\frac{(\mu_2 - \mu_1)^2}{\sigma_2^2} + \log\frac{\sigma_1}{\sigma_2} - \frac{1}{2}.$$

- Two multivariate Gaussians $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ and $q(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$,

$$D_{\mathrm{KL}}[p\|q] = \frac{1}{2}\mathrm{tr}(\boldsymbol{\Sigma}_2^{-1}\boldsymbol{\Sigma}_1) + \frac{1}{2}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)^\top \boldsymbol{\Sigma}_2^{-1}(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1) - \frac{1}{2}D + \frac{1}{2}\log\frac{|\boldsymbol{\Sigma}_2|}{|\boldsymbol{\Sigma}_1|}.$$

Calculus

Calculus

- Extrema
- Gradients and the Jacobian
- The Hessian
- Matrix calculus
- Important rules for differentiation
- One example of integration and approximate integration

Extrema

- Optimization is about finding extrema.
- Suppose $f : \mathbb{R}^d \to \mathbb{R}$. A point $\mathbf{x}$ is said to be a local minimum of f in $\mathcal{X} \subseteq \mathbb{R}^d$ if $f(\mathbf{x}) \leq f(\mathbf{y})$ for all $\mathbf{y}$ in some neighborhood $N \subseteq \mathcal{X}$ about $\mathbf{x}$.
- If $f(\mathbf{x}) \leq f(\mathbf{y})$ for all $\mathbf{y} \in \mathcal{X}$, then $\mathbf{x}$ is a global minimum of f in $\mathcal{X}$.

Gradients and the Jacobian

- The gradient of $f : \mathbb{R}^d \to \mathbb{R}$, denoted ∇f is given by

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}.$$

- The Jacobian of $f : \mathbb{R}^n \to \mathbb{R}^m$ is a matrix of first-order partial derivatives:

$$\mathbf{J}_f = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

The Hessian

- The Hessian of $f : \mathbb{R}^d \to \mathbb{R}$ is a matrix of second-order partial derivatives:

$$\nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_d \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_d^2} \end{bmatrix}.$$

Matrix Calculus

- Two examples of matrix calculus are

$$\nabla_{\mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \mathbf{a}$$
$$\nabla_{\mathbf{x}}(\mathbf{x}^\top \mathbf{A}\mathbf{x}) = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}.$$

- See Matrix Cookbook for the details.

Important Rules for Differentiation

- Sum rule is

$$(f(x) + g(x))' = f'(x) + g'(x) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}.$$

- Product rule is

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x) = \frac{\partial f}{\partial x}g(x) + f(x)\frac{\partial g}{\partial x}.$$

- Chain rule is

$$(g \circ f)'(x) = (g(f(x)))' = g'(f(x))f'(x) = \frac{\partial g}{\partial f}\frac{\partial f}{\partial x}.$$

One Example of Integration and Approximate Integration

- Expectation is computed by integration:

$$\mathbb{E}[f(\mathbf{x})] = \int f(\mathbf{x})p(\mathbf{x})\ \mathrm{d}\mathbf{x}.$$

- Because it is intractable in practice, approximate integration is needed. One of approximate integrations, Monte Carlo method is

$$\mathbb{E}[f(\mathbf{x})] = \int f(\mathbf{x})p(\mathbf{x})\ \mathrm{d}\mathbf{x} \approx \frac{1}{S}\sum_{i=1}^{S} f(\mathbf{x}^{(i)})$$

where $\mathbf{x}^{(i)} \sim p(\mathbf{x})$.

Revisit: Multivariate Gaussian Distribution

- Multivariate Gaussian distribution is

$$
\begin{aligned}
p(\mathbf{x}) &= \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) \\
&= \frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}.
\end{aligned}
$$

- Cumulative distribution function is

$$
F_{\mathbf{X}}(\mathbf{x}) = \int_{-\infty}^{\mathbf{x}} p(\mathbf{t})\, \mathrm{d}\mathbf{t} \approx \frac{1}{S}\sum_{\mathbf{s}\in\mathcal{X}} 1
$$

where $\mathcal{X} = \{\mathbf{s}|\mathbf{s} \leq \mathbf{x}, \mathbf{s} \sim p(\mathbf{x})\}$.

Thank you.

References

- Professor Seungjin Choi's materials
- https://gwthomas.github.io/docs/math4ml.pdf
- https://www.doc.ic.ac.uk/~dfg/ProbabilisticInference/Mathematics%20for%20machine%20learning.pdf